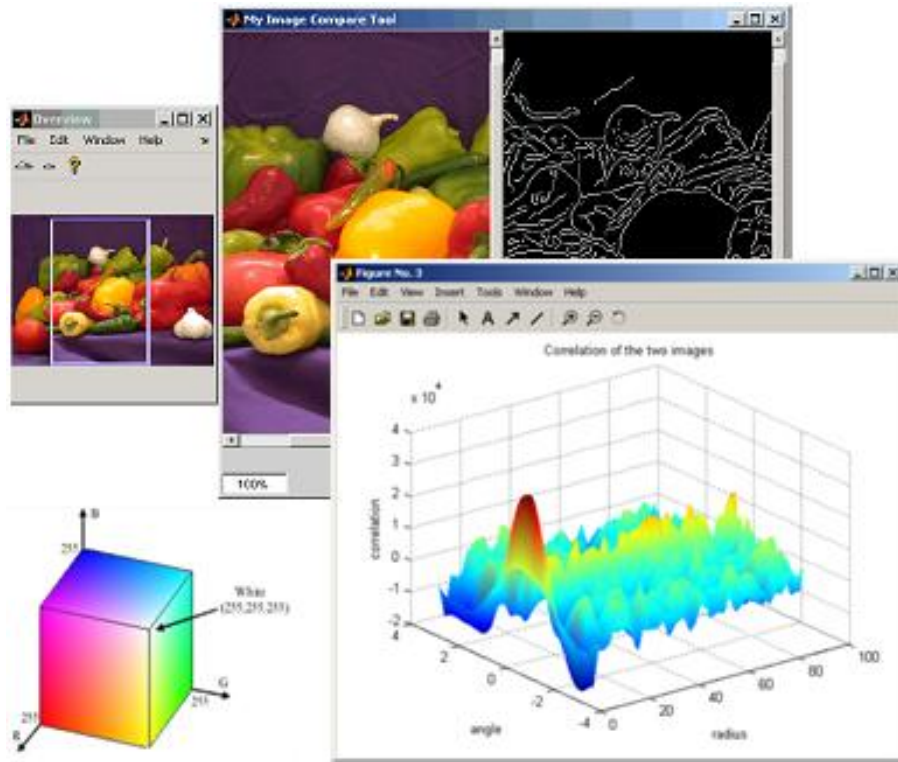


Digital Image Processing

LECTURE 11: PRACTICE SESSION



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Histogram Equalisation

Numerical Example

TYPE 2

Ques1: Apply the histogram Equalization technique to the following image

$f(x, y) =$

1	2	3	4
5	5	6	6
6	7	6	6
6	7	2	3

Numerical Example

Solution:

r_k	n_k	$\text{pdf} = \frac{n_k}{N}$	$\text{cpdf} (\Sigma)$	$(L-1)\Sigma$	o/p gray level (s_k)
0	0	0	0	0	0
1	1	0.06	0.06	0.42	0
2	2	0.13	0.19	1.33	1
3	2	0.13	0.32	2.24	2
4	1	0.06	0.38	2.66	3
5	2	0.13	0.51	3.57	4
6	6	0.38	0.89	6.23	6
7	2	0.13	1.02	7.14	7
Total (N) = 16					

Numerical Example

Solution:

i/p gray level (r_k)	o/p gray level (s_k)
0	0
1	0
2	1
3	2
4	3
5	4
6	6
7	7

$f(x, y) =$

1	2	3	4
5	5	6	6
6	7	6	6
6	7	2	3

$g(x, y) =$

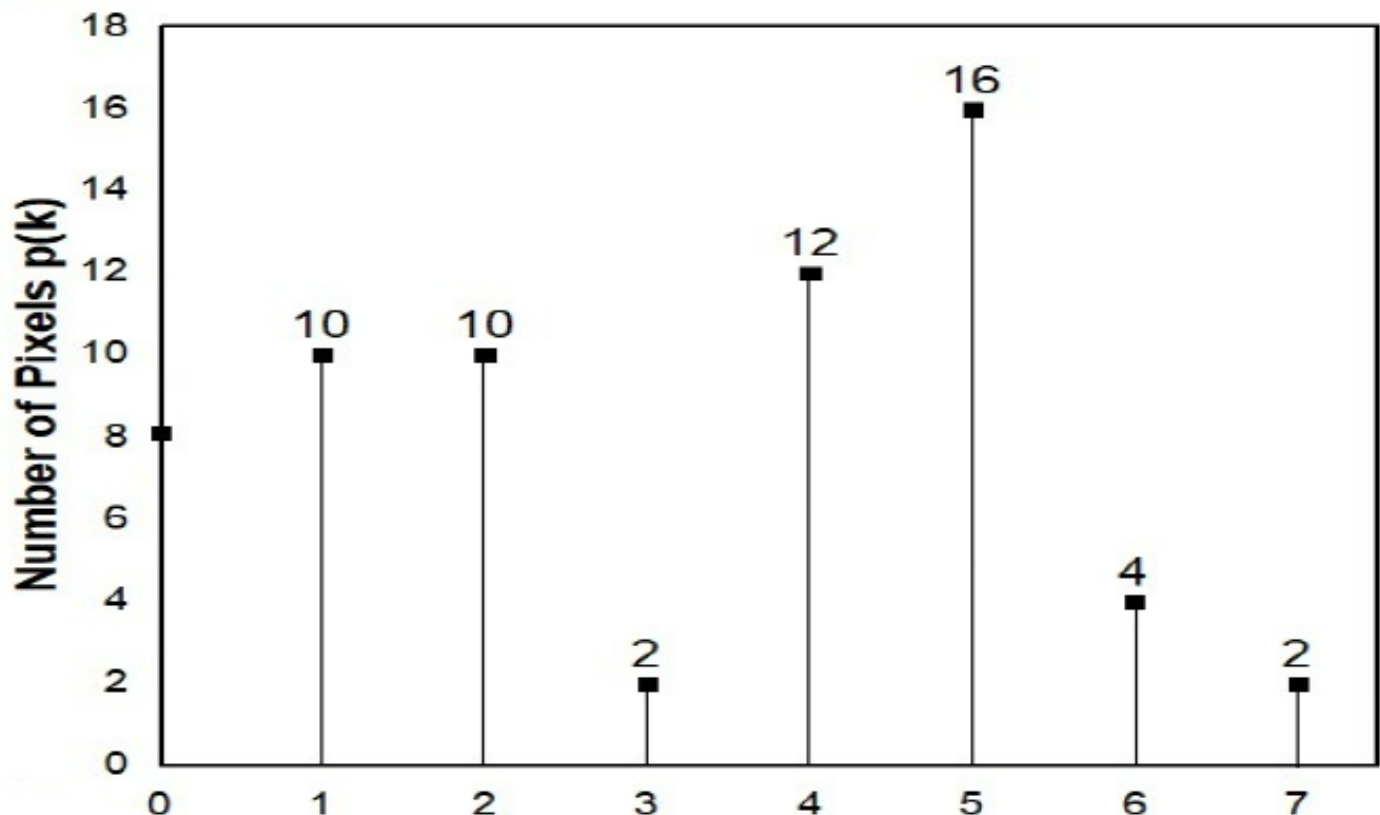
0	1	2	3
4	4	6	6
6	7	6	6
6	7	1	2

Numerical Examples

Ques 8: Perform Histogram Equalisation for 8×8 eight level image described in Figure. Also, construct the histogram of the equalized image.

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TYPE 3



Numerical Example

Solution:

r_k	n_k	$\text{pdf} = \frac{n_k}{N}$	$\text{cpdf} (\Sigma)$	$(L-1)\Sigma$	o/p gray level (s_k)
0	8	0.13	0.13	0.91	1
1	10	0.16	0.29	2.03	2
2	10	0.16	0.45	3.15	3
3	2	0.03	0.48	3.36	3
4	12	0.19	0.67	4.69	5
5	16	0.25	0.92	6.44	6
6	4	0.06	0.98	6.86	7
7	2	0.03	1.01	7.07	7
Total (N) = 64					

Numerical Example

Equalised (S_k)	No of Pixels (N_k)
0	0
1	8
2	10
3	12
4	0
5	12
6	16
7	6

Histogram Specification (Matching)

Numerical Examples

Ques1: Suppose that a 3-bit image of size 64×64 pixels has the intensity distribution shown in the table (on the left). Get the histogram transformation function and make the output image with the specified histogram, listed in the table on the right

Table 1

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

Table 2

z_q	Specified $p_z(z_q)$
$z_0 = 0$	0.00
$z_1 = 1$	0.00
$z_2 = 2$	0.00
$z_3 = 3$	0.15
$z_4 = 4$	0.20
$z_5 = 5$	0.30
$z_6 = 6$	0.20
$z_7 = 7$	0.15

TYPE 2

Numerical Examples

Solution:

Step1: Perform the Histogram Equalization for TABLE 1.

(r_k)	(n_k)	$p(r_k)$	Σ	$(L-1)\Sigma$	(s_k)
0	790	0.19	0.19	1.33	1
1	1023	0.25	0.44	3.08	3
2	850	0.21	0.65	4.55	5
3	656	0.16	0.81	5.67	6
4	329	0.08	0.89	6.23	6
5	245	0.06	0.95	6.65	7
6	122	0.03	0.98	6.86	7
7	81	0.02	1.00	7.00	7

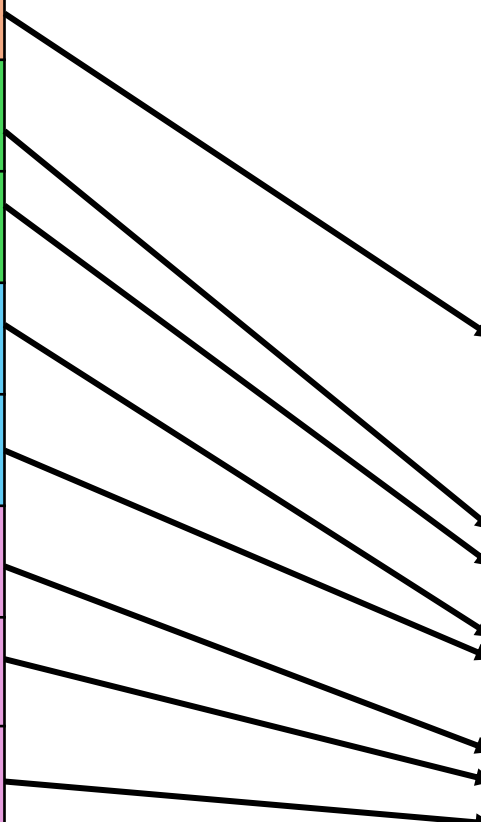
Numerical Examples

Step2: Perform the Histogram Equalization for TABLE 2.

(z_k)	$p(z_k)$	Σ	$(L-1)\Sigma$	$G(z_k)$
0	0.00	0	0	0
1	0.00	0	0	0
2	0.00	0	0	0
3	0.15	0.15	1.05	1
4	0.20	0.35	2.45	2
5	0.30	0.65	4.55	5
6	0.20	0.85	5.95	6
7	0.15	1	7	7

Numerical Examples

Step 3: For each gray level k , find s_k and then find a level such that S_k **Matching Criteria:** $G_{zK} - S_k \geq 0$ for min value of Z_q

r_k	s_k		G_{zK}	Z_q
0	1		0	0
1	3		0	1
2	5		0	2
3	6		1	3
4	6		2	4
5	7		5	5
6	7		6	6
7	7		7	7

Numerical Examples

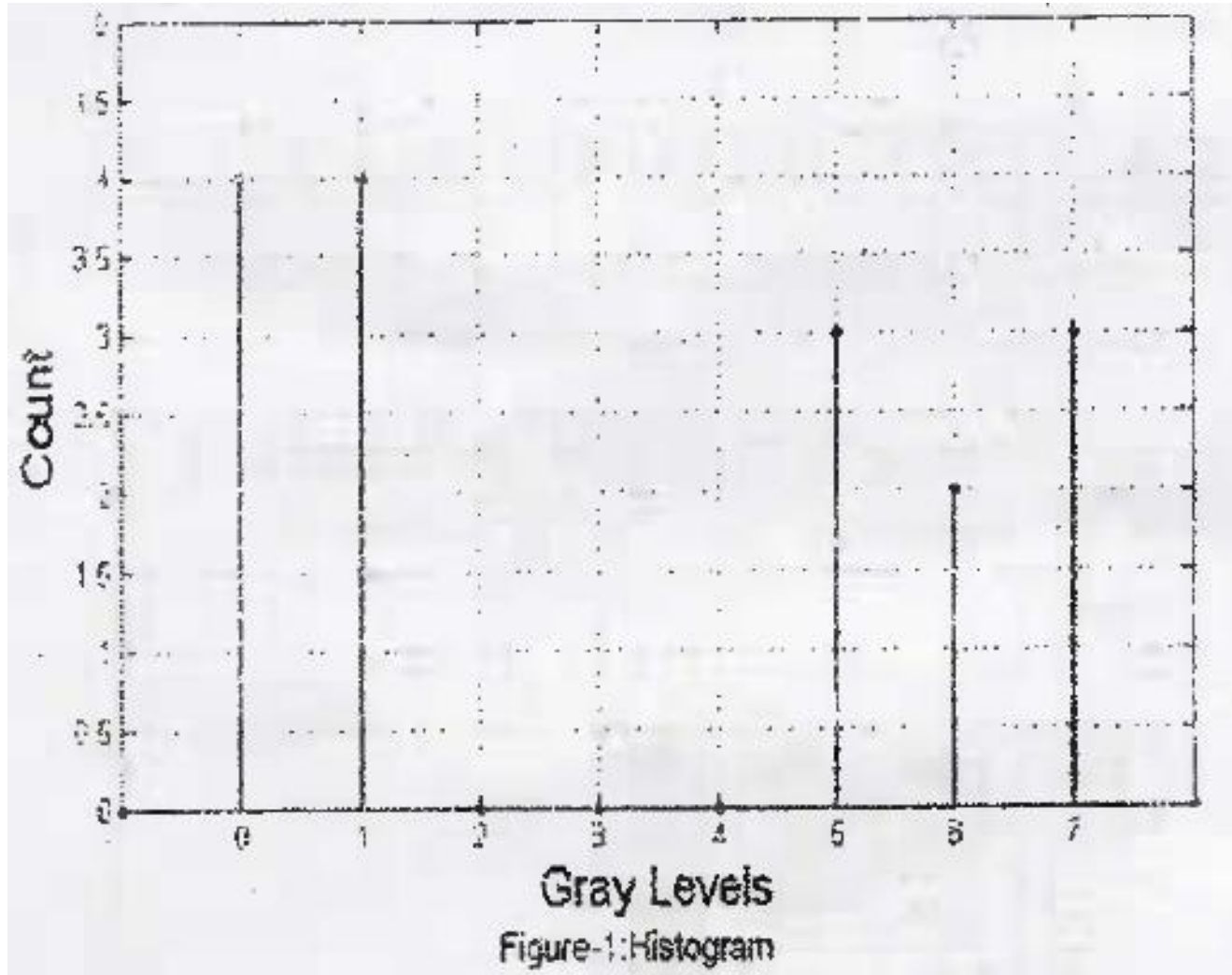
Step 4: Final Mapping

r_k	s_k	G_{zk}	Mapping
0	1	0	3
1	3	0	5
2	5	0	5
3	6	1	6
4	6	2	6
5	7	5	7
6	7	6	7
7	7	7	7

Numerical Examples

Ques 3: Histogram of a 3-bits/pixel square image is given in figure 1,.

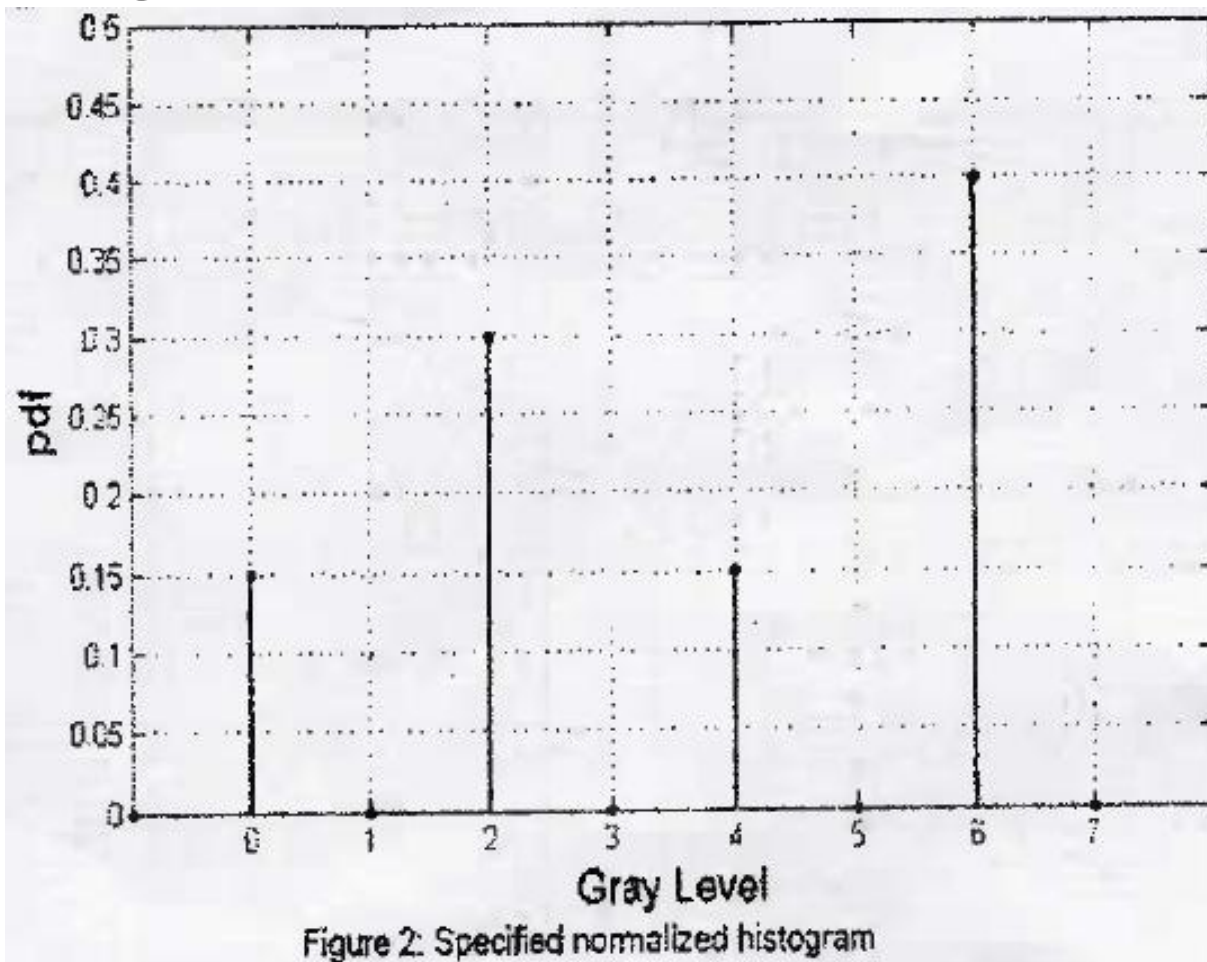
TYPE 3



Numerical Examples

specified normalised histogram (pdf) is given in figure 2.
Use histogram specification to generate histogram of enhanced image

TYPE 3



Practice Questions

Numerical Examples

Ques1: A and B are two 3-bit image. Histogram Equalisation was done on both the images and following results were obtained.

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I/P gray level of Image A	0	1	2	3	4	5	6	7
O/P gray level of Image A	1	3	5	6	6	7	7	7

I/P gray level of Image A	0	1	2	3	4	5	6	7
O/P gray level of Image A	0	0	0	1	2	5	6	7

Numerical Examples

We want to match the histogram of Image A to that of Image B. What will be the result of histogram matching? Give your answer in the following format.

I/P gray level of Image A	0	1	2	3	4	5	6	7
O/P gray level of Image A after matching								

Numerical Question

Ques 2: You are given 4×4 image with 2 bit per pixel. Can this image be enhanced using histogram Equalisation.

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1	1	2	2
3	3	0	0
2	1	1	2
3	0	3	0

Numerical Examples

Ques 3: A mask is defined with the following mathematical expression:

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$$f(x-2,y) + 2f(x-1, y) + 2f(x+1, y) + f(x+2, y) + f(x, y-2) + 2f(x, y-1) + 2f(x, y+1) + f(x, y+2) - 12f(x, y).$$

Draw this mask.

Numerical Examples

Ques 4: if $X = \{2 \ 3 \ 4 \ 3 \ 4 \ 5 \ 6\}$, and $W = \{1 \ \underline{1} \ 1\}$. What would be the result of median filtering? Use zero padding if required.

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Numerical Examples

Ques 5: Consider an image shown in Fig.5 (alias I6). Suppose we apply a 1×3 average filter on an image I6 to get the image I7. Again we apply the same filter on image I7 to get the image I8.

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2	5	6	6	7	4	3
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Find the final value of the center pixel.

Numerical Examples

Ques 6: Perform the log transformation on the image shown in fig. 1. Assume the values of C equals to the number of planes in an image.

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19	8
2	5

Numerical Examples

Ques 7: For the following given image

1	2	3	4
5	5	6	6
6	7	6	6
6	7	2	3

Assuming the gray level is 0 to 7 i.e. 8, determine the following

- a) Inversion/digital negative operation $g(x, y) = L-1-f(x, y)$
- b) Square function $g(x, y) = f(x, y)^2$
- c) Logarithmic operation $g(x, y) = 0.5*\log[f(x, y)]+1$
- d) Power law operation $g(x, y) = 1*f(x, y)^{1.2}$

Numerical Examples

Ques8: Suppose there is an image of dimension 5x5 in which 3 bits are used to represent a pixel. How many distinct 3-bit images can be generated in 5x5 grid.

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$$8^{25}$$

Numerical Examples

Ques 9: Consider the image as shown in fig 2. Convert this image in a binary image by applying threshold, which is the mean of this image.

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0	0	0	4	6
3	3	3	2	1
5	5	3	2	2
2	2	0	1	1
4	4	3	1	3

Numerical Examples

Ques 10: A 7-bit digital image has a histogram where the gray levels are equally distributed in the range from 70 to 110. For each operation of the following transformation functions, describe the gray levels range in which the pixels will lie. Also draw the transformation function for each.

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- Image negative
- Addition of 40 to all pixels gray levels
- Application of thresholding functions where the threshold is selected as gray level 95.

Numerical Examples

Ques 11: You have 4×5 image as follows:

2	3	5	5	6
5	3	9	11	12
12	11	1	8	1
7	7	9	0	1

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Perform Gray level Slicing with respect to given function – clearly mention the new pixel values in 4×5 matrix. Also, Represent the function graphically.

$$S = \begin{cases} 15 & \text{if } 5 < r \leq 10 \\ r & \text{otherwise} \end{cases}$$



Thank You